Muhammad Muizz Zafar

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Technical Skills

Programming Languages:	C++, C#, Python, Java, JavaScript, TypeScript, GoLang, SQL, HTML/CSS
Frameworks/Tools:	NUnit, Google Test, Cypress, Jenkins, AWS, Docker, Vue.js, React, Angular, Git

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work Experience

Software & Database Analyst (Data Engineering & Analytics) — May 2024 - Present
Elections Canada — Ottawa, ON

- Developed a reusable ETL validation framework using **SQL (T-SQL)**, **Python** and **Pandas**, enabling automated validation of large datasets (**20M+ records**) across dev, uat, and prod and created visual dashboards for data quality KPIs using **Streamlit**
- Built and maintained scalable **ETL pipelines** using **Python**, **PySpark**, **Pandas**, and **SQL** to extract data from **Oracle** databases, transform it, and load it into semi-structured data in a **JSON data lake** and MSSQL data warehouse, processing **5GB+** of data per pipeline run.
- Managed **MSSQL** database ETL releases and coordinated version-controlled deployment scripts, ensuring traceability, and rollback capability
- Assisted with **database monitoring and maintenance**, including performance diagnostics, log analysis, and troubleshooting database issues across enterprise systems.
- Worked in a cross-functional **Agile team** collaborating with business stakeholders, database administrators, and technical teams to define data validation requirements and improve reporting accuracy.

Full Stack Developer — September 2023 - August 2024
University of Ottawa — Ottawa, ON

- Collaborated with engineering teams in an **Agile sprint cycle** to review test results, ensure feature reliability, and fix UI/API integration bugs.
- Optimized SQL queries and LINQ REST APIs in C#, achieving 15-second faster data lookups across the event system.
- Revamped the event registration and admin pages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code
- Redesigned the University of Ottawa's emergency alert website with **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily. **website: ue.uottawa.ca/**
- Optimized existing **.Net Core REST APIs** by reducing SQL query execution time for event data lookup by 15 seconds
- Wrote 9 **End to End tests** using **Cypress**, and **25 unit tests** using **Jasmine** for new APIs and features implemented on frontend and **NUnit** backend

Cloud Developer Intern — Sep 2022 - Dec 2022 & May 2023 - Aug 2023
Trend Micro Inc. — Ottawa, ON

- Built automation for AWS-based testing environments using **GoLang**, **AWS Lambda**, and **Athena**, improving test data processing efficiency.
- Configured **CloudWatch** and **Splunk** alarms to proactively detect errors, reducing false test results and downtime.
- Developed and maintained automated integration tests for serverless pipelines to ensure stability of continuous releases.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Automated event detection workflows by integrating **Python**, **PostgreSQL**, and **GraphQL**, reducing manual QA verification time.
- Designed real-time dashboards in **Vue.js** for better monitoring and quicker validation of production events.

Full Stack Developer — June 2019 - June 2021
Tech Under Twenty — Oakville, ON

- Created and tested RESTful APIs using **.NET Core (C#)** for the member management system, improving stability and data validation accuracy.
- Designed responsive UI components in **Angular**, performing manual testing and debugging to improve user experience.